LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Equipment setup, tools and spares for Arc welding and procedure
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Equipment setup, tools and spares for Arc welding and procedure
5. Specific Objectives	31.1.4.1 Trainee will be able to explain arc welding and brazing procedure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Arc welding equipment setup, tool identification, and safety checks; Instructor supervision of safety gear donning and circuit verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an arc welding machine?
(b) List three essential safety precautions before starting arc welding.
(c) Identify the two main types of electrodes used in arc welding.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Arc welding machine components • Electrode holder and ground clamp setup • Cable routing and connection checks • Power source selection and settings • Workpiece preparation and cleaning	L & Ds	PPT, IB	15 min
(b)	• Essential tools for arc welding • Spares inventory and maintenance • Electrode storage and handling • Personal protective equipment (PPE) usage • Fire extinguisher and first aid kit location	L & Ds	PPT, PDF	20 min
(c)	• Arc welding procedure steps • Brazing procedure overview • Quality control checks during welding • Common defects and prevention methods • Post-welding cleanup and inspection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the importance of proper electrode storage in arc welding.
(b) Describe the key differences between arc welding and brazing procedures.
(c) List the critical steps in post-welding inspection to ensure quality.

12. Home Assignment and Follow Up	Prepare a brief report on the tools and spares required for arc welding, referencing the master syllabus for LP 11.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Performing Butt/Lap joint by Arc welding procedure and Inspection of welded job	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min